

MINGZE LI

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SUMMARY

Student Research Assistant at *FZI Forschungszentrum Informatik* and M.Sc. graduate from *KIT*, focusing on **deep learning for medical and scientific imaging**.

I build data-centric, reproducible pipelines for (i) scale-aware skin/psoriasis lesion segmentation and area quantification from smartphone images, and (ii) retinal OCT motion-artifact correction with 3D U-Net + CBAM on large volumes (manuscript in preparation).

Research focus: reliability-focused imaging ML—robust generalization under limited/shifted data and careful evaluation.

RESEARCH INTERESTS

- **Deep learning for medical & scientific imaging:** segmentation, restoration, 3D/volumetric learning
- **Data-centric ML:** dataset curation, augmentation strategies, label/noise robustness, evaluation design
- **Robustness & generalization:** domain shift, external validation, robustness analysis in real-world
- **Uncertainty & reliability:** calibration, confidence estimation, QC and failure-mode analysis
- **Reproducible research engineering:** clean pipelines, ablations, metric-driven reporting, GPU training

PUBLICATIONS

- [1]. Konle, J., Li, M., et al. Working title: **Motion-artifact correction in retinal OCT with 3D U-Net + CBAM (in preparation)**. Target venue (tentative): IEEE CISA 2026. (second author).
- [2]. Hofmann, J., Schweickart, L., Zimmermann, C., Li, M., Below, J., Schmieder, A., ... & Stork, W. (2025, June). **Perfusion Analysis for Telemedical Long-Term Monitoring of Patients with Chronic Skin Diseases**. In 2025 6th International Conference in Electronic Engineering & Information Technology (EEITE) (pp. 1-6). IEEE.
- [3]. Li, M., Yang, H., Liu, Z., Alam, M. M., Sack, H., & Gesese, G. A. (2024). **KGMistral: Towards boosting the performance of large language models for question answering with knowledge graph integration**. In Workshop on Deep Learning and Large Language Models for Knowledge Graphs.

COMPETITIONS

Kaggle RobustMedCT 2025 (Medical CT robustness challenge)

- Built ResNet18/34 pipelines with class-balanced sampling and corruption-aware augmentations for 11-class abdominal CT classification.
- 4-model probability ensemble: **0.797** public accuracy; ranked 19th on the public leaderboard.

EDUCATION

10/2020 – 04/2025 **Information Systems (M.Sc.) - Research focus: Machine Learning for Medical Imaging**
Karlsruhe Institute of Technology (KIT), Karlsruhe, Germany
Thesis: Designing an Iterative Prompt Optimization System for Domain-Specific Chatbots

09/2015 - 06/2019 **Information Management and Information Systems (B.Sc.)**
Qilu University of Technology, Jinan, China

Quantitative foundations: Calculus, Linear Algebra, Probability & Statistics, Operations Research; ML foundations

RESEARCH EXPERIENCE

07/2024 - Present **Student Research Assistant, FZI Forschungszentrum Informatik Karlsruhe**

OKTI Project [1] (07/2025 - Present)

- Constructed a controllable **simulated OCT dataset** by augmenting public volumes and injecting synthetic fixational eye-movement trajectories to create paired corrupted/clean training data.
- Implemented a **differentiable warping** pipeline with pixel-level lateral displacement labels and volume-level train/val/test splits.

- Trained a **3D U-Net + CBAM** multi-task model with a reconstruction head and a shift-estimation head for joint artifact correction and motion prediction.
- Designed losses combining **masked 3D SSIM + intensity** terms and an L_1 shift regression loss, with staged weighting to stabilize training.
- Built a reproducible PyTorch workflow (windowed 3D inputs, mixed precision, early stopping) and added QC audits (e.g., phase-correlation consistency checks) to improve supervision quality.

HybridVITA Project [2] (07/2024 - 06/2025)

- Curated and preprocessed smartphone dermatology images; implemented illumination-robust preprocessing (e.g., white balance) and data augmentation to improve generalization.
- Fine-tuned an FCN-ResNet50 (PyTorch) for skin/psoriasis segmentation for downstream lesion quantification, with ROI/background handling depending on the dataset; designed post-processing to obtain clinically usable masks.
- Integrated segmentation outputs into a telemedical longitudinal monitoring pipeline, enabling consistent region extraction for downstream analysis across sessions.
- Implemented an OpenCV ArUco-based module for scale calibration and lesion area quantification to support follow-up comparison in remote monitoring.
- Prepared analyses and communicated model behavior/failure cases to clinical collaborators; contributed technical content to the associated publication.

11/2024 – 04/2025 Master thesis: Designing an Iterative Prompt Optimization System for Domain-Specific Chatbots (H-Lab, KIT)

- Designed and implemented an iterative **multi-agent debating system** to refine prompts for a **mental health support chatbot**.
- Grounded the system in Design Science Research with design principles based on **Cognitive Load Theory** and Adaptive Structuration Theory.
- Automated synthetic dialogue generation and **GPTScore-based evaluation**, with domain experts in the loop via an interactive web interface.
- Conducted a mixed-methods **user study** (experts & lay users, **PHQ-9**, **NASA-TLX**, qualitative feedback) and derived prescriptive design implications

11/2023 – 03/2024 Engineering Information System - EIS Capstone Project (H-Lab, KIT)

- Worked with **multimodal data** (eye-tracking, HR, PPG, accelerometer, mouse clicks and movement) collected in a lab study on an ABB operator-training simulator to study workload and performance.
- Designed analyses and visual reports to give operators and supervisors feedback on **stress, attention, and performance**.
- Prototyped a **biosignal-adaptive smartwatch interface**, exploring how physiological data can be integrated into everyday work to support safer and more efficient operation.

11/2023 – 03/2024 Behavioral Activation Chatbot (H-Lab, KIT)

- Developed a **behavioral activation chatbot** using a **Retrieval-Augmented Generation (RAG)** pipeline and a **fine-tuned GPT-3.5** model.
- Implemented mood tracking, personalized activity recommendations, four incentive mechanisms, and schedule management via **Google Calendar API** to support sustained engagement.
- Ran a SurveyMonkey **user study** (n = 25) to evaluate usability, mood-tracking accuracy, and impact on BA, showing overall positive effects on users' self-management and well-being.

11/2023 – 03/2024 Boosting the Performance of Large Language Models for Question Answering with Knowledge Graph Integration [3] (KIT)

- Implemented entity/relation extraction from questions and **SPARQL** over domain **KG**.
- Verbalized relevant triples and processed prompt engineering.
- Evaluated with **SBERT** similarity, **ROUGE**, and **BLEU**.
- Improved factuality and grounding of our pipeline named **KG-Mistral** vs. other baselines

SKILLS

- **Deep learning for imaging:** PyTorch; segmentation & restoration; 3D/volumetric models
- **Medical CV:** OpenCV; preprocessing & augmentation; ArUco-based scale calibration
- **Robust evaluation:** domain shift; QC/failure analysis; uncertainty & calibration
- **Research engineering:** Python; Git; Linux GPU workflows; experiment tracking
- (Secondary) NLP: RAG; KG-based QA; prompt engineering